

TAREK BADIH

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EDUCATION

University of California, Berkeley

Class of 2026

Bachelor of Arts in Data Science

Relevant Coursework:

Data Structures, Foundations of Data Science, Principles & Techniques of Data Science, Introduction to Artificial Intelligence, The Structure and Interpretation of Computer Programs

EXPERIENCE

Data Science Intern

May 2024 – Aug 2024

BLCK UNICRN

Los Angeles, CA

- Developed a dynamic pricing model for BLCK UNICRN's VR experiences, increasing ticket sales by **12%** and improving accessibility. Optimized pricing via A/B testing, boosting retention by **18%**.
- Processed large datasets using **SQL** and **Python**, applying data cleaning and feature engineering. Built predictive models with **92%** accuracy to inform pricing and marketing.
- Developed predictive analytics pipelines for dynamic pricing, driving a **15%** revenue increase through optimized pricing and promotions.

PROJECTS

Sustainable Food Preservation System | Python, Arduino, TensorFlow

June 2022 – July 2022

- Designed an eco-friendly food storage solution for the **MIT Hackathon**, extending shelf life by **50%** using advanced sealing and preservation techniques.
- Developed a cost-effective, biodegradable container leveraging vacuum-sealing and moisture control to reduce spoilage in developing regions.
- Worked with multidisciplinary teams to source datasets, validate materials, and refine prototypes, ensuring scalability and environmental sustainability.

Bio-Inspired Transit Optimization | Python, GIS, Blender

Click me for link to project!

- Developed a bio-inspired transportation model for the **Biodesign Challenge 2024**, using slime mold simulations to optimize Berkeley's transit routes.
- Analyzed urban mobility data and integrated 3D city models with biological simulations to create efficient public transportation strategies.
- Collaborated across biology and data science disciplines, leveraging GIS mapping and computational modeling to enhance urban mobility solutions.

Build Your Own World | Java, Git, Randomized Algorithms

Jan. 2024

- Led a capstone project developing a video game engine that generates interactive 2D worlds using procedural generation and randomized algorithms.
- Engineered dynamic world generation, real-time user interaction via GUI, and implemented core features such as keyboard input handling, save/load functionality, and lighting mechanics.
- Developed logic for map boundaries, save-state persistence, and interactive game mechanics including reset and replay options.

Autonomous Drone Logistics System | Python, Algorithm Design, Embedded Systems

Nov. 2023

- Designed an autonomous drone delivery system for the **ENJAZ Tech Competition**, earning **3rd place** for its efficiency and scalability.
- Developed an optimized routing algorithm using real-time pathfinding techniques to enhance delivery speed and operational efficiency.
- Integrated software and hardware components, iterating on solutions to address technical challenges in a competitive environment.

SKILLS

Languages: Python, Java, SQL, C, R, HTML, JavaScript

Frameworks & Tools: Jupyter Notebook, Git, VS Code, IntelliJ, TensorFlow, PyTorch, Scikit-learn, Apache Spark

Technologies & Libraries: pandas, NumPy, Matplotlib, Seaborn, SciPy, OpenCV, SQLAlchemy